

Lecture Notes

Course: Complex Analysis

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Topic: Henstock-Kurzweil Integral

Lemma 8.1. If $x \in I$, there exists a subinterval $[x_i, x_i]$ from $\mathcal{P}^*$ that contains x . Since $\mathcal{P}^* \ll \delta$, then

$$x_i^* - \delta(x_i^*) \leq x_{i-1} \leq x \leq x_i \leq x_i^* + \delta(x_i^*)$$

so that $x_i^* - \delta(x_i^*) \leq x \leq x_i^* + \delta(x_i^*)$ which is equivalent to $|x - x_i^*| < \delta(x_i^*)$.

Theorem 8.3 (Cousin's Theorem). Let E denote the set of all points $x \in [a, b]$ such that there exists a δ -fine partition of the subinterval $[a, x]$. Given δ , the set E is nonempty, since for $x \in [a, a + \delta(a)]$ we have that the pair $([a, x], a)$ is a δ -fine partition because for $p \in [a, x]$ we have $p \leq x \leq a + \delta(a)$ and $a - \delta(a) < a \leq p$ because $\delta(a) > 0$, so we conclude that $[a, x] \subset [a, a + \delta(a)]$. Moreover, given that $E \subseteq [a, b]$, E is bounded and thus it has a supremum, call it $u \doteq \sup E$ which is such that $u \leq b$.

Since $u - \delta(u) < u = \sup E$, there exists $\nu \in E$ such that $u - \delta(u) < \nu < u$, in this case, given that $\nu \in E$, we have that there exists a δ -fine partition $\mathcal{P}_1^*$ of $[a, \nu]$ and if we let $\mathcal{P}_2^* \doteq \mathcal{P}_1^* \cup ([\nu, u], u)$, then any subinterval is either in $\mathcal{P}_1^*$ (and thus satisfy the condition for $\mathcal{P}^*$ to be δ -fine) or is $[\nu, u]$, but given that $u - \delta(u) < \nu$ and that $u < u + \delta(u)$ since $\delta(u) > 0$, we conclude that this subinterval also satisfies the required condition, so that $\mathcal{P}_2^*$ is a δ -fine partition (of $[a, u]$) and thus $u \in E$.

Suppose $u < b$, then let $w \in [a, b]$ be such that $u < w < u + \delta(u)$. If $\mathcal{Q}_1^*$ is a δ -fine partition of $[a, u]$ (which exists since we showed that $u \in E$), then let $\mathcal{Q}_2^* \doteq \mathcal{Q}_1^* \cup ([u, w], u)$ so that by a similar argument as in i) we get that $\mathcal{Q}_2^*$ is a δ -fine partition of $[a, w]$, whence $w \in E$, a **contradiction**, since $u < w$ and $u \in E$, so u is not an upper bound of E .

Theorem 8.4 (Uniqueness Theorem). Suppose that L_1, L_2 are two numbers satisfying the definition, then we have that for any given $\varepsilon > 0$, there exist gauges $\delta'_{\varepsilon/2}$ and $\delta''_{\varepsilon/2}$ such that

$$|S(f; \mathcal{P}^*) - L_1| < \frac{\varepsilon}{2} \quad \text{and} \quad |S(f; \mathcal{P}^*) - L_2| < \frac{\varepsilon}{2}$$

provided that $\mathcal{P}^* \ll \delta'_{\varepsilon/2}$ and $\mathcal{P}^* \ll \delta''_{\varepsilon/2}$, respectively. Hence, choose $\delta_\varepsilon = \inf\{\delta'_{\varepsilon/2}, \delta''_{\varepsilon/2}\}$ which is a gauge on I , and thus for $\mathcal{P}^* \ll \delta_\varepsilon$ we have

$$|L_1 - L_2| \leq |S(f; \mathcal{P}^*) - L_1| + |S(f; \mathcal{P}^*) - L_2| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

The conclusion now immediately follows from the arbitrariness of ε .

Theorem 8.6. Let $g \in \mathcal{HK}[a, b]$ and suppose that f and g differ only at one point in $[a, b]$, say c , then

$$|S(f; \mathcal{P}^*) - L| \leq |S(f; \mathcal{P}^*) - S(g; \mathcal{P}^*)| + |S(g; \mathcal{P}^*) - L| \quad (*)$$

but the worst case is that c is a partition point and the tag of each of the adjacent subintervals, in this case $c \in I_k, I_{k+1}$ and the Riemann sums of f and g differ at most at these two terms so that

$$\begin{aligned} |S(f; \mathcal{P}^*) - S(g; \mathcal{P}^*)| &= |f(c)\ell(I_k) + f(c)\ell(I_{k+1}) - g(c)\ell(I_k) - g(c)\ell(I_{k+1})| \\ &\leq |f(c)\ell(I_k)| + |f(c)\ell(I_{k+1})| + |g(c)\ell(I_k)| + |g(c)\ell(I_{k+1})|, \end{aligned}$$

but if $\mathcal{P}^* \ll \delta_\varepsilon$ then $I_k, I_{k+1} \subseteq \bar{V}_{\delta}(c)$ and thus

$$|S(f; \mathcal{P}^*) - S(g; \mathcal{P}^*)| \leq (|f(c)| + |g(c)|)8\delta(c). \quad (**)$$

Define

$$\delta_\varepsilon(x) = \begin{cases} \delta_{\varepsilon/2}^g & x \in [a, b] \setminus \{c\}, \\ \inf\{\delta_{\varepsilon/2}^g, \frac{\varepsilon}{(17(|f(c)| + |g(c)|))}\} & x = c, \end{cases}$$

where $\delta_{\varepsilon/2}^g$ is the gauge needed to make $|S(g; \mathcal{P}^*) - L| < \varepsilon/2$ so that if $\mathcal{P}^* \ll \delta_\varepsilon$, then by (*) and (**) we conclude that

$$|S(f; \mathcal{P}^*) - L| < \varepsilon,$$

provided that $\mathcal{P}^* \ll \delta_\varepsilon$. So $f \in \mathcal{HK}[a, b]$ with integral L .

Theorem 8.7 (Properties of the Henstock-Kurzweil Integral). i) We show the case for real f, g and complex k and then let $f = f_1 + \sqrt{-1}f_2, g = g_1 + \sqrt{-1}g_2$. Let f_1, g_1 be real and $\mathbb{C} \ni k \neq 0$ (the case $k = 0$ is trivial and treated separately), then there exists a gauge δ_ε on $[a, b]$ such that if $\mathcal{P} \ll \delta_\varepsilon$ is a partition of $[a, b]$, then

$$\left| S(f; \dot{\mathcal{P}}) - \int_a^b f \right| < \frac{\varepsilon}{2|k|} \quad \text{and} \quad \left| S(g; \dot{\mathcal{P}}) - \int_a^b g \right| < \frac{\varepsilon}{2},$$

and since $S(kf + g; \dot{\mathcal{P}}) = kS(f; \dot{\mathcal{P}}) + S(g; \dot{\mathcal{P}})$, we get that

$$\left| S(kf + g; \dot{\mathcal{P}}) - \left(k \int_a^b f + \int_a^b g \right) \right| \leq \left| kS(f; \dot{\mathcal{P}}) - k \int_a^b f \right| + \left| S(g; \dot{\mathcal{P}}) - \int_a^b g \right| < \frac{|k|\varepsilon}{2|k|} + \frac{\varepsilon}{2} = \varepsilon$$

ii) Suppose f, g are real and such that $f \leq g$ on I and that $\int_a^b f > \int_a^b g$. Then given any starred partition $\mathcal{P}^*$, we have

$$S(f; \mathcal{P}^*) = \sum_{i=1}^n f(x_i^*)(x_i - x_{i-1}) \leq \sum_{i=1}^n g(x_i^*)(x_i - x_{i-1}) = S(g; \mathcal{P}^*),$$

because $x_i^* \in I$ for all $i = 1, \dots, n$, and then by integrability we know that given $\varepsilon > 0$, there exists a $\delta_{\varepsilon/2}$ such that if $\mathcal{P}^* \ll \delta_{\varepsilon/2}$, then

$$\int_a^b f - \frac{\varepsilon}{2} < S(f; \mathcal{P}^*) \quad \text{and} \quad S(g; \mathcal{P}^*) < \int_a^b g + \frac{\varepsilon}{2}$$

which implies that

$$\int_a^b f - \int_a^b g < \varepsilon,$$

but since $\varepsilon > 0$ is arbitrary we can choose $\varepsilon = \frac{1}{2} \left(\int_a^b f - \int_a^b g \right) > 0$ so that

$$\int_a^b f - \int_a^b g < \frac{1}{2} \left(\int_a^b f - \int_a^b g \right) \iff 1 < \frac{1}{2},$$

clearly, a contradiction.

Theorem 8.10 (Additivity Theorem). ($\Rightarrow$) Let f_1, f_2 be the restrictions of f to $[a, c]$ and $[c, b]$, respectively and suppose that they are both integrable to L_1, L_2 . This means that given $\varepsilon > 0$, there exist gauges δ' and δ'' of $[a, c]$ and $[c, b]$ such that

$$|S(f_1; \mathcal{P}_1^*)| < \frac{\varepsilon}{2} \quad \text{and} \quad |S(f_2; \mathcal{P}_2^*)| < \frac{\varepsilon}{2}$$

for $\mathcal{P}_1^* \ll \delta'$ and $\mathcal{P}_2^* \ll \delta''$. Let us now define a gauge δ_ε on $[a, b]$ by

$$\delta_\varepsilon(t) \doteq \begin{cases} \inf\{\delta'(t), \frac{1}{2}(c-t)\} & t \in [a, c), \\ \inf\{\delta'(c), \delta''(c)\} & t = c, \\ \inf\{\delta''(t), \frac{1}{2}(t-c)\} & t \in (c, b]. \end{cases}$$

We show that this gauge has the property that any subinterval I of a δ_ε -fine partition containing c must have c as its tag. For this, suppose I_i is a subinterval with tag t of a starred partition that is

δ_ε -fine and suppose $t < c$, then (recall that any m -fine partition with $m = \inf\{\gamma_1, \gamma_2\}$ is also γ_1 - and γ_2 -fine)

$$[x_{i-1}, x_i] \subseteq [t - \frac{1}{2}(c-t), t + \frac{1}{2}(c-t)],$$

but suppose for the sake of contradiction, that $c \in [t - (c-t)/2, t + (c-t)/2] = [3t/2 - c/2, t/2 + c/2]$ which is equivalent to

$$3t/2 - c/2 \leq c \leq t/2 + c/2 \iff 3t - c \leq 2c \leq t + c \implies c \leq t,$$

clearly, a contradiction. The same goes for $t > c$ in which we must have

$$t - \frac{1}{2}(t-c) \leq c \leq t + \frac{1}{2}(t-c) \iff \frac{t}{2} + \frac{c}{2} \leq c \leq \frac{3t}{2} - \frac{c}{2} \implies t \leq c,$$

again, a contradiction. We therefore conclude that the only way for c to be in I_i is for c to be the tag thereof. We now show that any δ -fine partition of $[a, b]$, then there exist $\mathcal{Q}_1^* \ll \delta'$ and $\mathcal{Q}_2^* \ll \delta''$ partitions of $[a, c]$ and $[c, b]$, respectively, such that

$$S(f; \mathcal{Q}^*) = S(f_1; \mathcal{Q}_1^*) + S(f_2; \mathcal{Q}_2^*). \quad (*)$$

First, suppose c is a partition point of $\mathcal{Q}^*$ then it belongs to two subintervals I_j and I_{j+1} and thus it must be tag for both of these intervals. In this case let $\mathcal{Q}_1^*$ be the subintervals contained in $[a, c]$ and $\mathcal{Q}^*$ be the ones contained in $[c, b]$, it is clear from the definition of δ_ε that $\mathcal{Q}_1^*$ is δ' fine and similarly for $\mathcal{Q}_2^*$ with δ'' . If c is not a partition point, then it must be contained in some I_j , for which it is a tag thereof. Replace the pair $([x_{j-1}, x_j], c)$ by the two pairs $([x_{j-1}, c], c)$ and $([c, x_j], c)$ and let $\mathcal{Q}_1^*$ be the subintervals in $[a, c]$ for this new partition, similarly with $\mathcal{Q}_2^*$ and $[c, b]$. Notice that $f(c)(x_j - x_{j-1}) = f(c)(c - x_{j-1}) + f(c)(x_j - c)$. In each case we get that $(*)$ holds. In any case, by the triangle inequality we have

$$|S(f; \mathcal{Q}^*) - (L_1 + L_2)| \leq |S(f; \mathcal{Q}_1^*) - L_1| + |S(f; \mathcal{Q}_2^*) - L_2| < \varepsilon,$$

meaning that $L = L_1 + L_2$ with L the integral of f over $[a, b]$, i.e.,

$$\int_a^b f = \int_a^c f + \int_c^b f$$

Conversely, if $f \in \mathcal{HK}[a, b]$, then given $\varepsilon > 0$ let η_ε be the gauge satisfying the Cauchy criterion. Let f_1 be the restriction of f to $[a, c]$ and let $\mathcal{P}_1^* = \{I_k, x_k^*\}_{k=1}^n$, $\mathcal{Q}_1^* = \{J_k, y_k^*\}$ be η_ε -fine partitions of $[a, c]$. We add partition points p_j, q_j in $[c, b]$ to extend $\mathcal{P}_1^*$ and $\mathcal{Q}_1^*$ to partitions $\mathcal{P}^*$ and $\mathcal{Q}^*$ of $[a, b]$. Notice that if we let $p_j = q_j$ for the extension, then

$$S(f; \mathcal{P}^*) - S(f; \mathcal{Q}^*) = S(f_1; \mathcal{P}_1^*) - S(f_1; \mathcal{Q}_1^*),$$

meaning that

$$|S(f; \mathcal{P}_1^*) - S(f; \mathcal{Q}_1^*)| < \varepsilon,$$

because $|S(f; \mathcal{P}^*) - S(f; \mathcal{Q}^*)| < \varepsilon$, but since $\mathcal{P}_1^*, \mathcal{Q}_1^*$ are any η_ε -fine partitions, then the restriction f_1 belongs to $\mathcal{HK}[a, c]$. Similarly, $f_2 \in \mathcal{HK}[c, b]$. Given that both f_1 and f_2 are HK-integrable, we let L_1, L_2 be their respective integrals and thus by the first part of the proof we have that if f was HK-integrable to L then $L = L_1 + L_2$, i.e.,

$$\int_a^b f = \int_a^c f_1 + \int_c^b f_2 = \int_a^c f + \int_c^b f.$$